

Practical: 4	Implementation and Time analysis of factorial program using iterative and recursive method
18/08/2026	

Aim: To Implement and Time analysis of factorial program using iterative and recursive method.

Algorithm:

Algorithm

1) Iterative method:

Input: user input number
 output: factorial of the input number

step-1:- Initialize fact = 1
 step-2:- If condition of
 if (num == 0)
 step-3:- for i = num in condition i >= 1
 then do
 fact = fact * i;
 step-4:- decrement of i as i--;
 step-5:- return fact
 step-6:- stop

2) Recursive Method:-

Input: user input number
 output:- factorial number of input number

step-1:- start initialize fact = 1
 step-2:- If condition of
 if (num == 0)
 step-3:- while (num >= 1)
 fact = fact * num;
 num--
 fact_recursive (num);
 step-4:- return fact;
 step-5:- stop

Program:

factorial program using iterative and recursive method

```
#include<iostream>
using namespace std;
```

```
int fact_iterative(int num){
    int fact = 1;
    if(num == 0)
        return 1;
    for(int i=num; i>=1; i--){
        fact = fact*i;
    }
    return fact;
}

int fact_recursive(int num){
    int fact = 1;
    if(num == 0)
        return 1;
    while(num >= 1){
        fact = fact*num;
        num--;
        fact_recursive(num);
    }
    return fact;
}

int main(){
    cout<<"-----Factorial-----"<<endl;
    int n;
    cout<<"Enter your number: ";
    cin>>n;
    int i = fact_iterative(n);

    cout<<"Iterative method:"<<endl;
    cout<<n<<" factorial is "<<i;
    cout<<endl<<endl;
    int r = fact_recursive(n);
    cout<<"Recursive method:"<<endl;
    cout<<n<<" factorial is "<<r;

    return 0;
}
```

Output:

```

C:\Users\Kotha\OneDrive\D  X + v
-----Factorial-----
Enter your number: 5
Iterative method:
5 factorial is 120

Recursive method:
5 factorial is 120
-----
Process exited after 2.87 seconds with return value 0
Press any key to continue . . . |

```

Time Complexity:

time complexity :

1. Iterative Method :

```

fact = 1;      → 1
if (num == 0); → 1
return 1       → 1
for (i = num; i >= 1; i--) { → 1 + n + 1 + n
    fact = fact * i; → n
}
return fact     → 1

```

$T(n) = 2n + 2 + n$
 $T(n) = 3n + 2$
 $T(n) = n$
 the time complexity is $O(n)$

Best case : $O(n)$
 Average : $O(n)$
 Worst : $O(n)$

2) Recursive method :

```

int fact (int num) { → 1
    int fact = 1; → 1
    if (num == 0) → 1
    return → 1
    while (num >= 1) { → n
        fact = fact * num; → n
        num--; → n
        fact (num); → 1
    }
    return fact; → 1

```

$T(n) = 3n + 4$
 $= n$
 Time complexity is $O(n)$
 space complexity is $O(1)$

Best case : $O(n)$
 Average : $O(n)$
 worst : $O(n)$

Conclusion:

The implementation of this program and the execution is successfully got executed and the time complexity in the Three cases are $O(n)$ in the both Iterative, recursive methods And Space complexity of both methods are $O(1)$.